

37MARA

**Outdoor Unit Single Zone Ductless System
Sizes 9 to 36**

INSTALLATION MANUAL

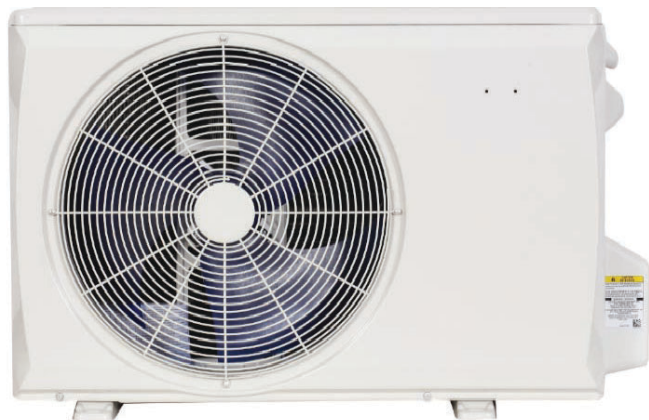


Fig. 1 — Sizes 06K to 36K

NOTES: Read the entire instruction manual before starting the installation.

Images are for illustration purposes only. Actual models may differ slightly.

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SAFETY CONSIDERATIONS

Installing, starting up, and servicing air- conditioning equipment can be hazardous due to system pressures, electrical components, and equipment location (roofs, elevated structures, etc.).

Only trained, qualified installers and service mechanics should install, start- up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as coil cleaning. All other operations should be performed by trained service personnel only.

When working on the equipment, observe the precautions in the literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and a fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read these instructions thoroughly and follow all warnings or cautions included in the literature and attached to the unit. Consult local building codes and National Electrical Code (NEC) for special requirements. Recognize safety information.

This is the safety - alert symbol .

When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury. Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**. These words are used with the safety- alert symbol.

DANGER identifies the most serious hazards which results in severe personal injury or death.

WARNING signifies hazards which could result in personal injury or death.

CAUTION is used to identify unsafe practices which may result in minor personal injury or product and property damage.

NOTE is used to highlight suggestions which results in enhanced installation, reliability, or operation.

R-454B		Refrigerant Safety Group A2L	R-454B
WARNING - Risk of Fire due to Flammable Refrigerant Used. Follow Handling Instructions Carefully in Compliance with National Regulations			
			349807-101 REV.-

NOTE: Risk of Fire. Flammable refrigerant used. To be repaired only by trained service personnel. Do not puncture refrigerant tubing.

	WARNING
ELECTRICAL SHOCK HAZARD Failure to follow this warning could result in personal injury or death. Before installing, modifying, or servicing system, the main electrical disconnect switch must be in the OFF position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.	

	WARNING
	EXPLOSION HAZARD Failure to follow this warning could result in death, serious personal injury, and/or property damage. Never use air or gases containing oxygen for leak testing or operating refrigerant compressors. Pressurized mixtures of air or gases containing oxygen can lead to an explosion.

	WARNING
PERSONAL INJURY AND PROPERTY DAMAGE HAZARD For continued performance, reliability, and safety, the only approved accessories and replacement parts are those specified by the equipment manufacturer. The use of non-manufacturer approved parts and accessories could invalidate the equipment limited warranty and result in a fire risk, equipment malfunction, and failure. Please review the manufacturer's instructions and replacement parts catalogs available from your equipment supplier.	

	CAUTION
EQUIPMENT DAMAGE HAZARD Failure to follow this caution may result in equipment damage or improper operation. Do not bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units. If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during extended periods of system shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.	

1. Installation (where refrigerant pipes are allowed)

Any person who is involved with working on or breaking into a refrigerant circuit should hold a current valid certificate from an industry-accredited assessment authority, which authorizes their competence to handle refrigerants safely in accordance with an industry recognized assessment specification.

Maintenance and repair requiring the assistance of other skilled personnel shall be carried out under the supervision of the person competent in the use of flammable refrigerants.

The installation of pipe-work shall be kept to a minimum.

Pipe-work shall be protected from physical damage.

Where refrigerant pipes shall be compliance with national gas regulations. That mechanical connections shall be accessible for maintenance purposes.

Be more careful that foreign matter (oil, water, etc.) does not enter the piping. When storing the piping, securely seal the opening by pinching, taping, etc. All working procedure that affects safety means shall only be carried by competent persons.

Appliance shall be stored in a well ventilated area where the room size corresponds to the room area as specific for operation.

Joints shall be tested with detection equipment with a capability of 1/8 oz (5g)/year of refrigerant or better, with the equipment in standstill and under operation or under a pressure of at least these standstill or operation conditions after installation. In cases that require mechanical ventilation, ventilation openings shall be kept clear of obstruction.

LEAK DETECTION SYSTEM installed. Unit must be powered except for service. For the unit with refrigerant sensor, when the refrigerant sensor detects refrigerant leakage, the indoor unit displays a error code and emit a buzzing sound, the compressor of outdoor unit immediately stops, and the indoor fan starts running. The service life of the refrigerant sensor is 15 years. When the refrigerant sensor malfunctions, the indoor unit displays the error code **FHCC**. **Refer to the error code table in the unit's service manual for details.** The refrigerant sensor can not be repaired and can only be replaced by the manufacture. It shall only be replaced with the sensor specified by the manufacture.

2. Because a **FLAMMABLE REFRIGERANT** is used, the requirements for installation space of appliance and/or ventilation requirements are determined according to:
 - the refrigerant charge used in the appliance,
 - the installation location,
 - the type of ventilation of the location or of the appliance.
 - piping material, pipe routing, and installation shall include protection from physical damage in operation and service, and be in compliance with national and local codes and standards, such as ASHRAE 15, IAPMO Uniform Mechanical Code, ICC International Mechanical Code, or CSA B52. All field joints shall be accessible for inspection prior to being covered or enclosed.
 - that protection devices, piping, and fittings shall be protected as far as possible against adverse environmental effects, for example, the danger of water collecting and freezing in relief pipes or the accumulation of dirt and debris;
 - that piping in refrigeration systems is designed and installed to minimize the likelihood of hydraulic shock damaging the system;
 - that steel pipes and components shall be protected against corrosion with a rustproof coating before applying any insulation;
 - that precautions shall be taken to avoid excessive vibration or pulsation;

- the minimum floor area of the room shall be mentioned in the form of a table or a single figure without reference to a formula;
- after completion of field piping for split systems, the field pipe-work shall be pressure tested with an inert gas and then vacuum tested prior to refrigerant charging, according to the following requirements:

- a. The required nitrogen test pressure is 500 PSI.
 - b. The test pressure after removal of pressure source shall be maintained for at least 1 hour with no decrease of pressure indicated by the test gauge, with test gauge resolution not exceeding 5% of the test pressure.
 - c. During the evacuation test, after achieving a vacuum level specified in the manual or less, the refrigeration system shall be isolated from the vacuum pump and the pressure shall not rise above 1500 microns within 10 min. The vacuum pressure level shall be specified in the manual, and shall be the lessor of 500 microns or the value required for compliance with national and local codes and standards, which may vary between residential, commercial, and industrial buildings.
- Field-made refrigerant joints indoors shall be tightness tested according to the following requirements: The test method shall have a sensitivity of 1/8 oz (5 g) per year of refrigerant or better under a pressure of at least 125% of the maximum allowable pressure. No leak shall be detected.

3. Qualification of Workers

Any maintenance, service and repair operations must be performed by skilled and authorized personnel. Every working procedure that affects safety means shall only be carried out by competent persons that joined the training and achieved competence should be documented by a certificate. The training of these procedures is carried out by national training organizations or manufacturers that are accredited to teach the relevant national competency standards that may be set in legislation. Examples for such working procedures are:

- breaking into the refrigerating circuit;
- opening of sealed components;
- opening of ventilated enclosures.

4. Ventilated Area

Ensure that the area is in the open or that it is adequately ventilated before breaking into the system or conducting any hot work. A degree of ventilation shall continue during the period that the work is carried out. The ventilation should safely disperse any released refrigerant and preferably expel it externally into the atmosphere.

5. Cabling

Check that cabling is not subjected to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects. The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

6. Detection of Flammable Refrigerants

Under no circumstances shall potential sources of ignition be used in the searching for or detection of refrigerant leaks. A halide torch (or any other detector using a naked flame) shall not be used. The following leak detection methods are deemed acceptable for refrigerant systems. Electronic leak detectors that have a sensitivity of 1/8 Oz (5g)/year may be used to detect leaks of flammable refrigerants. (Detection equipment shall be calibrated in a refrigerant free area.) Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used. Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed, and the appropriate percentage of gas (25% maximum) is confirmed. Leak detection fluids are also suitable for use in external leak detection.

NOTE: Examples of leak detection fluids are

- bubble method,
- fluorescent method agents.

If a leak is suspected, all open flames shall be removed/extinguished. If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak. See the following instructions of removal of refrigerant.

7. Evacuation

When breaking into the refrigerant circuit to make repairs - or for any other purpose conventional procedures shall be used. However, for flammable refrigerants it is important that best practice be followed, since flammability is a consideration. The following procedure shall be adhered to:

- safely recover refrigerant following local and national regulations;
- evacuate;
- purge the circuit with NITROGEN;
- evacuate (requirement);
- continuously flush or purge with NITROGEN when using flame to open circuit; and
- open the circuit

The refrigerant charge shall be recovered into the correct recovery cylinders. Charging must be performed by the liquid charging method. For appliances containing flammable refrigerants, the system shall be purged with oxygen-free nitrogen to render the appliance safe for flammable refrigerants. This process might need to be repeated several times. Compressed air or oxygen shall not be used for purging refrigerant systems.

For appliances containing flammable refrigerants, refrigerants purging shall be achieved by breaking the vacuum in the system with oxygen-free nitrogen and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum (requirement). This process shall be repeated until no refrigerant is within the system (requirement). When the final oxygen-free nitrogen charge is used, the system shall be vented down to atmospheric pressure to enable work to take place. The outlet for the vacuum pump shall not be close to any potential ignition sources, and ventilation shall be available.

Recovery: When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.

When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed. Ensure that the correct number of cylinders for holding the total system charge is available. All cylinders to be used are designated.

8. Charging Procedures

In addition to conventional charging procedures, the following requirements shall be followed:

Works shall be undertaken with appropriate tools only (in case of uncertainty, consult the manufacturer of the tools for use with flammable refrigerants)

Ensure that contamination of different refrigerants does not occur when using charging equipment. Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them. Charging must be performed by the liquid charging method.

Ensure that the refrigeration system is grounded prior to charging the system with refrigerant.

Label the system when charging is complete (if not already).

Extreme care shall be taken not to overfill the refrigeration system.

Prior to recharging the system it shall be pressure tested with oxygen free nitrogen (OFN). The system shall be leak tested on completion of charging but prior to commissioning. A follow up leak test shall be carried out prior to leaving the site.

Table 1 — A (min) Table**hinst: Height Above Floor Level to Center of Indoor Unit / feet (meters)**

MC or Mrel Refrigerant Charge Amount / pounds (kilograms)		6.0 (1.8)	6.5 (2.0)	7.0 (2.1)	7.5 (2.3)	8.0 (2.4)	8.5 (2.6)	9.0 (2.7)	9.5 (2.9)	10.0 (3.0.8)
	4.0 (1.8)	33 (3.1)	28 (2.6)	24 (2.2)	21 (1.9)	18 (1.7)	16 (1.5)	14 (1.3)	13 (1.2)	12 (1.1)
	4.5 (2.0)	41 (3.8)	35 (3.3)	30 (2.8)	26 (2.5)	23 (2.2)	21 (1.9)	18 (1.7)	16 (1.5)	15 (1.4)
	5.0 (2.3)	51 (4.7)	43 (4.0)	37 (3.5)	33 (3.0)	29 (2.7)	25 (2.4)	23 (2.1)	20 (1.9)	18 (1.7)
	5.5 (2.5)	61 (5.7)	52 (4.9)	45 (4.2)	39 (3.7)	35 (3.2)	31 (2.8)	27 (2.5)	24 (2.3)	22 (2.1)
	6.0 (2.7)	73 (6.8)	62 (5.8)	54 (5.0)	47 (4.4)	41 (3.8)	36 (3.4)	32 (3.0)	29 (2.7)	26 (2.4)
	6.5 (3.0)	86 (8.0)	73 (6.8)	63 (5.9)	55 (5.0)	48 (4.5)	43 (4.0)	38 (3.5)	34 (3.2)	31 (2.9)
	7.0 (3.2)	100 (9.3)	85 (7.9)	73 (6.8)	64 (5.9)	56 (5.2)	50 (4.6)	44 (4.1)	40 (3.7)	36 (3.3)
	7.5 (3.4)	114 (10.6)	97 (9.0)	84 (7.8)	73 (6.8)	64 (6.0)	57 (5.3)	51 (4.7)	46 (4.2)	41 (3.8)
	8.0 (3.6)	130 (12.1)	111 (10.3)	95 (8.9)	83 (7.7)	73 (6.8)	65 (6.0)	58 (5.4)	52 (4.8)	47 (4.4)
	8.5 (3.9)	147 (13.6)	125 (11.6)	108 (10.0)	94 (8.7)	83 (7.7)	73 (6.8)	65 (6.1)	59 (5.4)	53 (4.9)
	9.0 (4.1)	164 (15.3)	140 (13.1)	121 (11.3)	105 (9.8)	93 (8.6)	82 (7.6)	73 (6.8)	66 (6.1)	59 (5.5)
	9.5 (4.3)	183 (17.0)	156 (14.5)	135 (12.5)	117 (10.9)	103 (9.6)	91 (8.5)	81 (7.6)	73 (6.8)	66 (6.1)
	10.0 (4.5)	203 (18.9)	173 (16.1)	149 (13.9)	130 (12.1)	114 (10.6)	101 (9.4)	90 (8.4)	81 (7.5)	73 (6.8)

A-min: Required Minimum Room Area / Square Feet (Square Meters)**AREA FORMULA****Amin** is the required minimum room area in square feet/square meters**mc** is the actual refrigerant charge in the system in ft/kg**mREL** is the refrigerant releasable charge in ft/kg**hinst** is the height of the center of the appliance relative to the floor of the room after installation.**WARNING:** The minimum room area or minimum room area of conditioned space is based on a releasable charge or total system refrigerant charge.**For R454B refrigerant charge amount and minimum room area:**

The indoor unit should be installed at least 5.9ft/1.8m above the floor, and the minimum room area for operation or storage should be as specified in Table 1.

ACCESSORIES

The system is shipped with the following accessories. Use all of the installation parts and accessories to install the system. Improper installation may result in water leakage, electrical shock and fire, or cause the equipment to fail. Keep the installation manual in a safe place and do not discard any other accessories until the installation work has been completed.

Table 1 —Accessories

PART NO.	PART NAME	QTY.
1	Outdoor unit	1
-	Literature package including installation instructions and warranty	1
-	Mounting pads for the outdoor unit (helps with vibration prevention during unit operation)	4
-	Drain Joint	1
-	Drain Hose	1

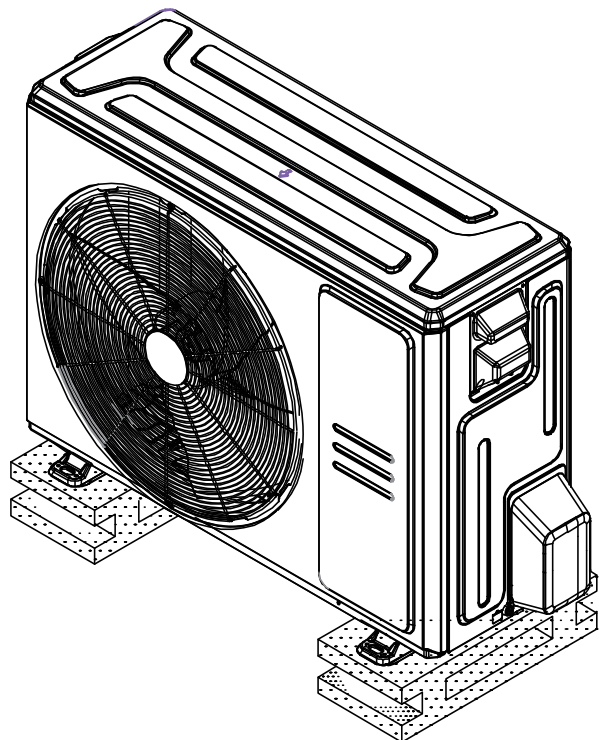


Fig. 2 — Outdoor Unit

NOTE:

- If the outdoor unit is higher than the indoor unit, prevent rain from flowing into the indoor unit along the connection pipe by making a downward arc in the connection pipe before it enters the wall to the indoor unit. This ensures that rain drips from the connection pipe before it enters the wall.

- Piping and the interconnecting wiring are field supplied.
- Figure 2 is only a sketch. The actual model may differ slightly.

Table 2 lists the units covered in this document.

Table 2 — Unit Sizes

SYSTEM TONS	BTUH	VOLTAGE	MODEL
1	12,000	115-1	37MARAQ12AA1
0.75	9,000	208/230-1	37MARAQ09AA3
1	12,000	208/230-1	37MARAQ12AA3
1.5	18,000	208/230-1	37MARAQ18AA3
2	24,000	208/230-1	37MARAQ24AA3
2.75	30,000	208/230-1	37MARAQ30AA3
3	36,000	208/230-1	37MARAQ36AA3

DIMENSIONS

Table 3 — Dimensions and Weights

SYSTEM SIZE		12K	9K	12K	18K	24K	30K	36K
		(115 V)	(208/230 V)	(208/230 V)	(208/230V)	(208/230 V)	(208/230 V)	(208/230 V)
Height (H)	in (mm)	21.85 (555)	21.85 (555)	21.85 (555)	26.50 (673)	31.89 (810)	31.89 (810)	31.89 (810)
Width (W)	in (mm)	30.12 (765)	30.12 (765)	30.12 (765)	35.04 (890)	37.24 (946)	37.24 (946)	37.24(946)
Depth (D)	in (mm)	11.93 (303)	11.93 (303)	11.93 (303)	13.46 (342)	16.14 (410)	16.14 (410)	16.14 (410)
Weight - Net	lbs. (kg)	64.6 (29.3)	62.17 (28.2)	62.17 (28.2)	92.59 (42)	122.14 (55.4)	142.86 (64.8)	150.36 (68.2)
PACKAGING								
Height (H)	in (mm)	24.02 (610)	24.02 (610)	24.02 (610)	29.13 (740)	34.84 (885)	34.84 (885)	34.84(885)
Width (W)	in (mm)	34.92 (887)	34.92 (887)	34.92 (887)	39.17 (995)	42.91(1090)	42.91(1090)	42.91(1090)
Depth (D)	in (mm)	13.27(337)	13.27(337)	13.27(337)	15.67(398)	19.69 (500)	19.69 (500)	19.69(500)
Weight - Gross	lbs. (kg)	70.11 (31.8)	67.68 (30.7)	67.68 (30.7)	100.31 (45.5)	132.28 (60)	154.54 (70.1)	160.28 (72.7)
Carton Drawing No.	--	ZXW-877*327* 590S-NSK(D)-K K-2	ZXW-877*327* 590S-NSK(D)-K K-2	ZXW-877*327* 590S-NSK(D)-K K-2	ZXW-985*388* 720S-NSK(D)-K K-2	ZXW-1075*485 *850S-NSK(D)- KK	ZXW-1075*485 *850S-NSK(D)- KK	ZXW-1075*485 *850S-NSK(D)- KK
Carton Material	--	Double Corrugated Cardboard						
Material Thick- ness	in (mm)	0.197 (5)	0.197 (5)	0.197 (5)	0.197 (5)	0.197 (5)	0.276 (7)	0.276 (7)

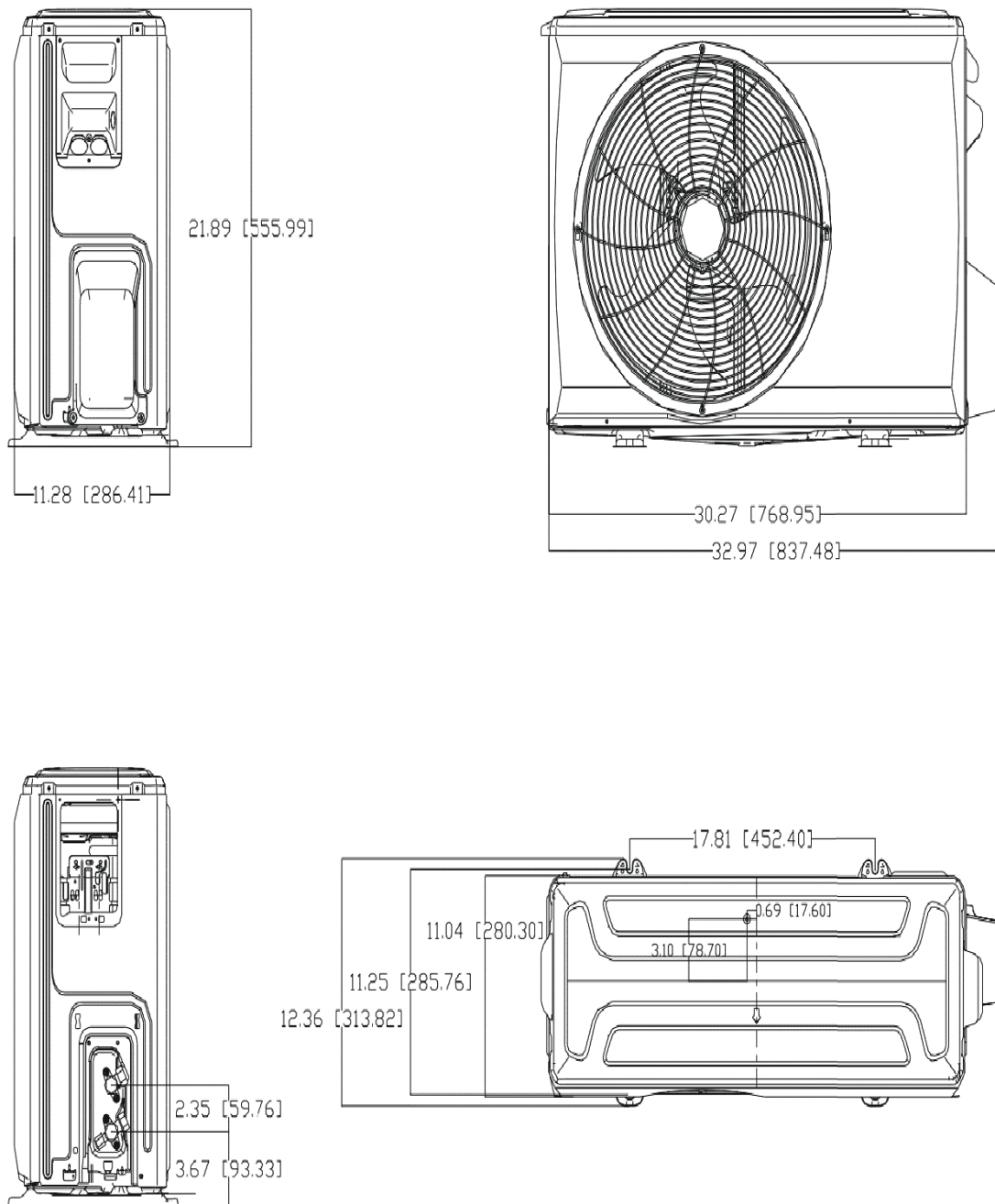


Fig. 3 — Sizes 06K/9K/12K

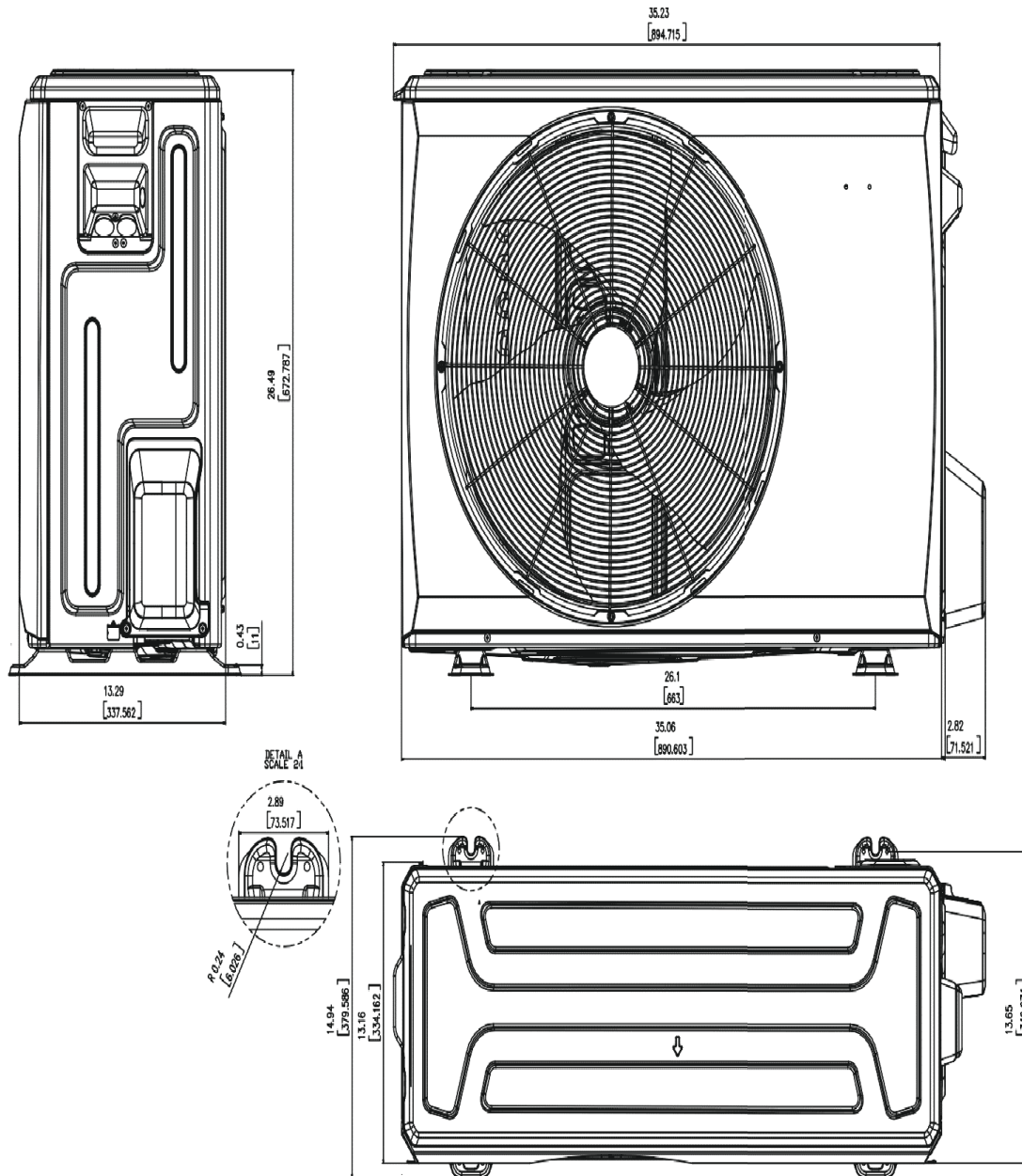


Fig. 4 — Size 18K

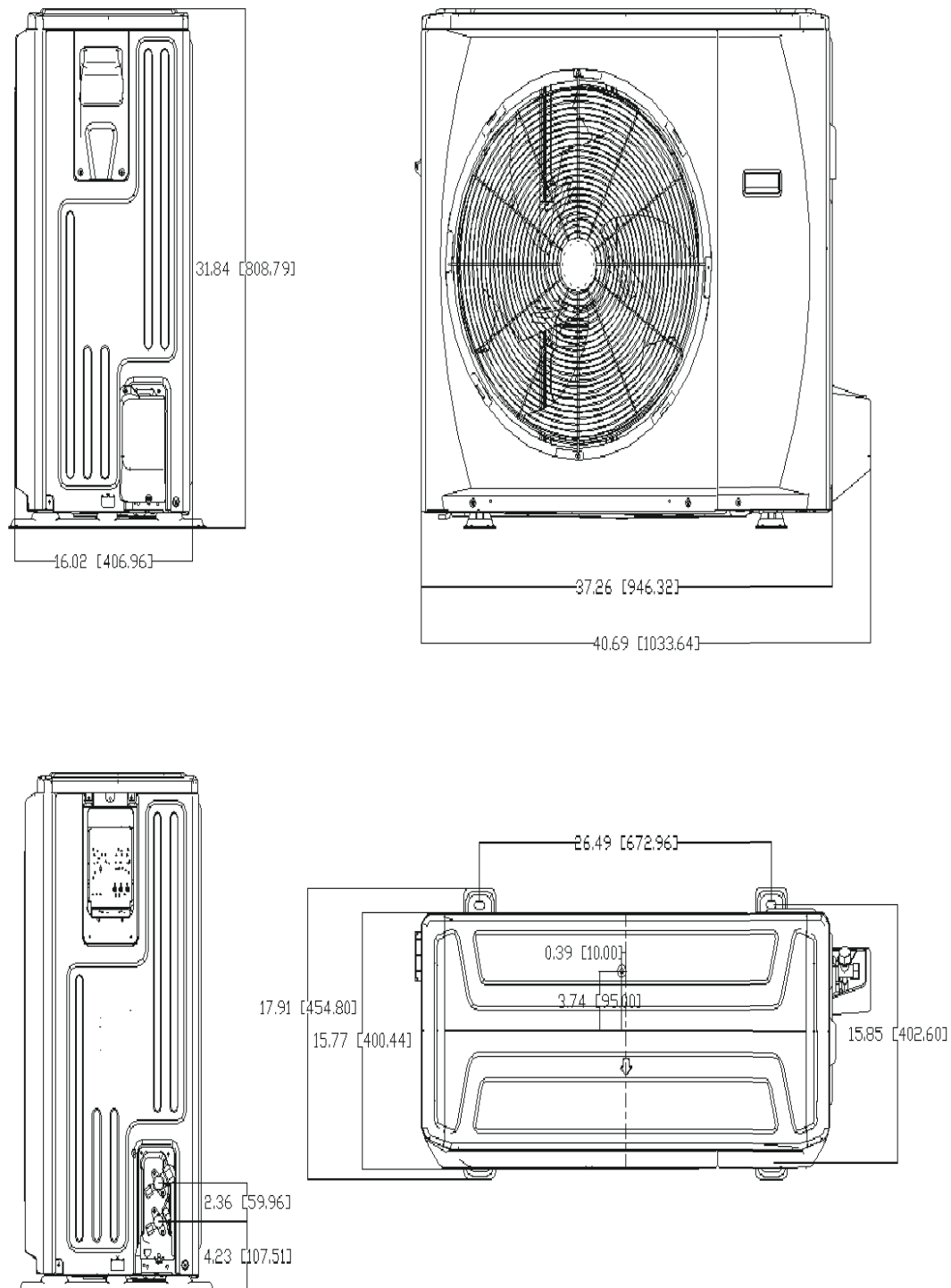


Fig. 5 — Sizes 24K/30K/36K

CLEARANCES

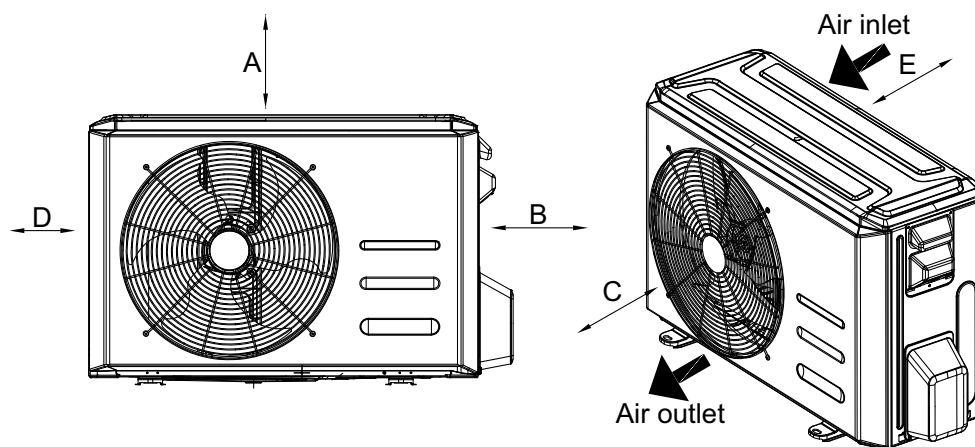


Fig. 6 — Clearances

Table 4 — Clearance Dimensions

UNIT	MINIMUM VALUE IN. (MM)
A	24 (610)
B	24 (610)
C	24 (610)
D	4 (101)
E	4 (101)

NOTE: The outdoor unit must be mounted at least 2in (50mm) above the maximum anticipated snow depth.

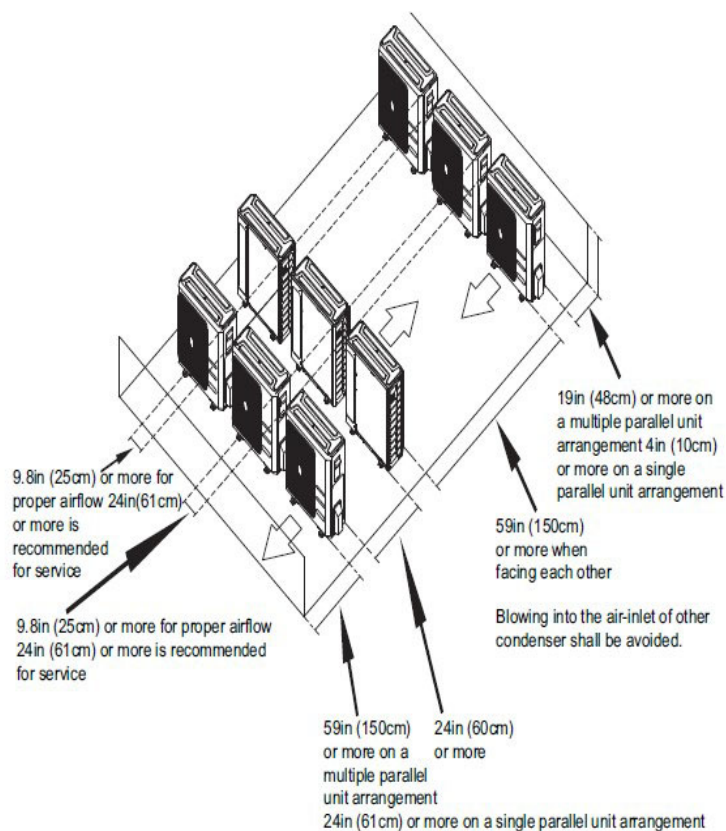


Fig. 7 — Clearances for multiple units

INSTALLATION

- A location which is convenient to installation and not exposed to strong winds.
- A location which can bear the weight of the outdoor unit and where the outdoor unit can be mounted in a level position.
- A location which provides appropriate clearances (see Fig. 6).
- Allow sufficient space for airflow and service of the unit. See Fig. 6 for the required minimum distances between the unit or walls.

NOTE: DO NOT install the indoor or outdoor units in a location with special environmental conditions. For those applications, contact your Ductless representative.



WARNING

PRODUCT INSTALLATION

Installation must be performed by an authorized dealer or specialist. A defective installation can cause water leakage, electrical shock, or fire.

The installation must be performed according to the installation instructions. Improper installation can cause water leakage, electrical shock, or fire. (In North America, installation must be performed in accordance with the requirements of NEC or CEC by authorized personnel only.)

Contact an authorized service technician for repair or maintenance of this unit. This appliance must be installed in accordance with local codes.

Only use the included accessories, parts, and specified parts for installation. Using non-standard parts can cause water leakage, electrical shock, fire, or unit failure.

To prevent exposure to wind, install the outdoor unit with its air inlet side facing the wall

Install drainage piping according to the instructions in this manual. Improper drainage may cause water damage to your home and property.

DO NOT install the unit in a location that may be exposed to combustible gas leaks. If combustible gas accumulates around the unit, it may cause a fire.

DO NOT turn on the power until all work has been completed.

When moving or relocating the system, consult experienced service technicians for the disconnection and re-installation of the unit.

NOTE: Base pan built-in with multiple holes for proper draining during defrost. For applications where it is required to seal these holes, and re-direct the condensate drain, rubber plugs are available through RCD.

Table 5 — Base Pan Rubber Plugs

UNIT	BASE PAN RUBBER PLUG PART NUMBER	QTY.
12k 115V	12600801A00077	30
9k	12600801A00077	30
12k	12600801A00077	30
18k	12600801A00077	30
24k	12600801A00117	5
30k	12600801A00117	5
36k	12600801A00117	5

Important: Both refrigerant lines must be insulated separately. Use refrigeration grade tubing ONLY. No other type of tubing may be used. Use of other types of tubing will void the manufacturer's warranty.

- The minimum refrigerant line length between the indoor and outdoor unit is 10 ft. (3m).
- When paired with Ductless indoor units, size the line sets based on the connection size of the indoor unit unless specified.
- When paired with conventional or multi-family fan coils, size the line sets based on the liquid and gas connection size of the outdoor unit; a field supplied reducer may be required.
- All outdoor units have an electronic expansion valve to manage the refrigerant flow of the fan coil connected.
- Do not open the service valves or remove the protective caps from the tubing ends until all connections are made.
- Bend the tubing with bending tools to avoid kinks and flat spots.
- Keep the tubing free of dirt, sand, moisture, and other contaminants to avoid damaging the refrigerant system.
- Avoid sags in the suction line to prevent the formation of oil traps.
- Insulate each tube with a minimum 3/8-in. (10 mm) wall thermal pipe insulation. Inserting the tubing into the insulation before making the connections saves time and improves installation quality.

R-454B



Refrigerant
Safety Group
A2L

R-454B

WARNING - Risk of Fire due to Flammable Refrigerant Used. Follow Handling Instructions Carefully in Compliance with National Regulations

349807-101 REV.-



WARNING

All field piping must be completed by a licensed technician and must comply with the local and national regulations. When the system is installed in a small room, measures must be taken to prevent the refrigerant concentration in the room from exceeding the safety limit in the event of refrigerant leakage. If the refrigerant leaks and its concentration exceeds its proper limit, hazards due to lack of oxygen may result. When installing the refrigeration system, ensure that air, dust, moisture or foreign substances do not enter the refrigerant circuit. Contamination in the system may cause poor operating capacity, high pressure in the refrigeration cycle, explosion or injury. Ventilate the area immediately if there is refrigerant leakage during the installation. Leaked refrigerant gas is hazardous. Ensure there is no refrigerant leakage after completing the installation work.



CAUTION

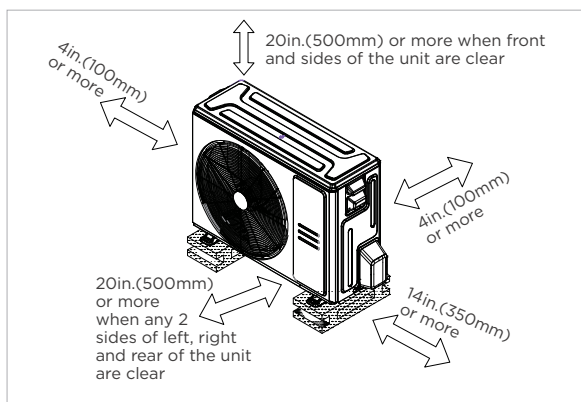
DO NOT install the connecting pipe until both the indoor and outdoor units have been installed. Insulate both the gas and liquid piping to prevent condensation.

REFRIGERANT PIPING**Table 6 — Refrigerant Piping Table**

OUTDOOR MODEL		37MARAQ12AA1	37MARAQ09AA3	37MARAQ12AA3	37MARAQ18AA3	37MARAQ24AA3	37MARAQ30AA3	37MARAQ36AA3
POWER SUPPLY	V;Ph;Hz	115V;1Ph;60HZ	208/230V;1Ph;60 HZ	208/230V;1Ph;60 HZ	208/230V;1Ph;60 HZ	208/230V;1Ph;60 HZ	208/230V;1Ph;60 HZ	208/230V;1Ph;60 HZ
PIPING AND REFRIGERANT INFORMATION								
Refrigerant Type	Type	R454B	R454B	R454B	R454B	R454B	R454B	R454B
Charge Amount	lb. (kg)	2.09(0.95)	2.03(0.92)	2.03(0.92)	3(1.36)	4.41(2.0)	5.29(2.4)	7.05(3.2)
Additional refrigerant charge	Oz/ft (g/m)	0.16(15)	0.16(15)	0.16(15)	0.16(15)	0.32(30)	0.32(30)	0.32(30)
Liquid Pipe (size - connection type)	In (mm)	6.35mm(1/4in)	6.35mm(1/4in)	6.35mm(1/4in)	6.35mm(1/4in)	9.52mm(3/8in)	9.52mm(3/8in)	9.52mm(3/8in)
Suction Pipe (size - connection type)	In (mm)	9.52mm(3/8in)	9.52mm(3/8in)	9.52mm(3/8in)	12.7mm(1/2in)	15.9mm(5/8in)	15.9mm(5/8in)	15.9mm(5/8in)
Min. Piping Length	ft. (m)	9.8 (3)	9.8 (3)	9.8 (3)	9.8 (3)	9.8 (3)	9.8 (3)	9.8 (3)
Standard Piping Length	ft. (m)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)
Max. Piping Length with no additional refrigerant charge per System	ft. (m)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)
Total Maximum Piping Length per system	ft. (m)	82.02(25)	82.02(25)	82.02(25)	98.42(30)	164.04(50)	164.04(50)	213.25(65)
Max. outdoor-indoor height difference (OU higher than IU)	ft. (m)	49.21(15)	49.21(15)	49.21(15)	65.62(20)	82.02(25)	82.02(25)	98.43(30)
Max. outdoor-indoor height difference (IU higher than OU)	ft. (m)	49.21(15)	49.21(15)	49.21(15)	65.62(20)	82.02(25)	82.02(25)	98.43(30)

A. Installation Instruction**Step One - Select the Installation Location**

NOTE: Before installing the outdoor unit, select an appropriate location. The following details are standards designed to help select an appropriate location for the unit.

Proper installation locations meet the following standards:

☒ Good air circulation and ventilation.



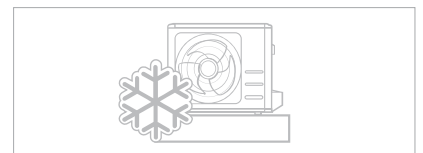
☒ Firm and solid—the location can support the unit and will not vibrate.



☒ Noise from the unit will not disturb other people.



☒ Protected from prolonged periods of direct sunlight or rain.



☒ Where snowfall is anticipated, take appropriate measures to prevent ice buildup and coil damage.

☒ Meets all spatial requirements shown in Installation Space Requirements above.

NOTE: Install the unit in accordance with the local codes and regulations.

⚠ CAUTION:**SPECIAL CONSIDERATIONS FOR EXTREME WEATHER**

If the unit is exposed to heavy wind:

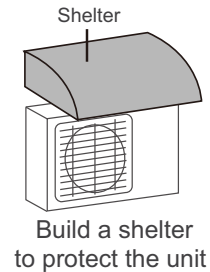
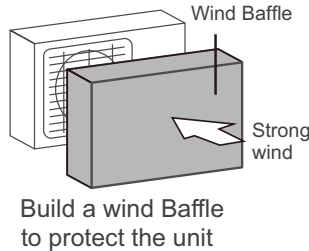
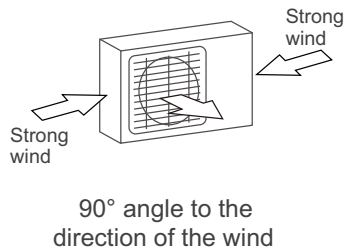
Install unit so the air outlet fan is at a 90° angle to the direction of the wind. If needed, build a barrier in front of the unit to protect it from extremely heavy winds. See Figures below.

If the unit is frequently exposed to heavy rain or snow:

Build a shelter above the unit to protect it from the rain or snow. Be careful not to obstruct air flow around the unit.

If the unit is frequently exposed to salty air (seaside):

Use outdoor unit that is specially designed to resist corrosion.

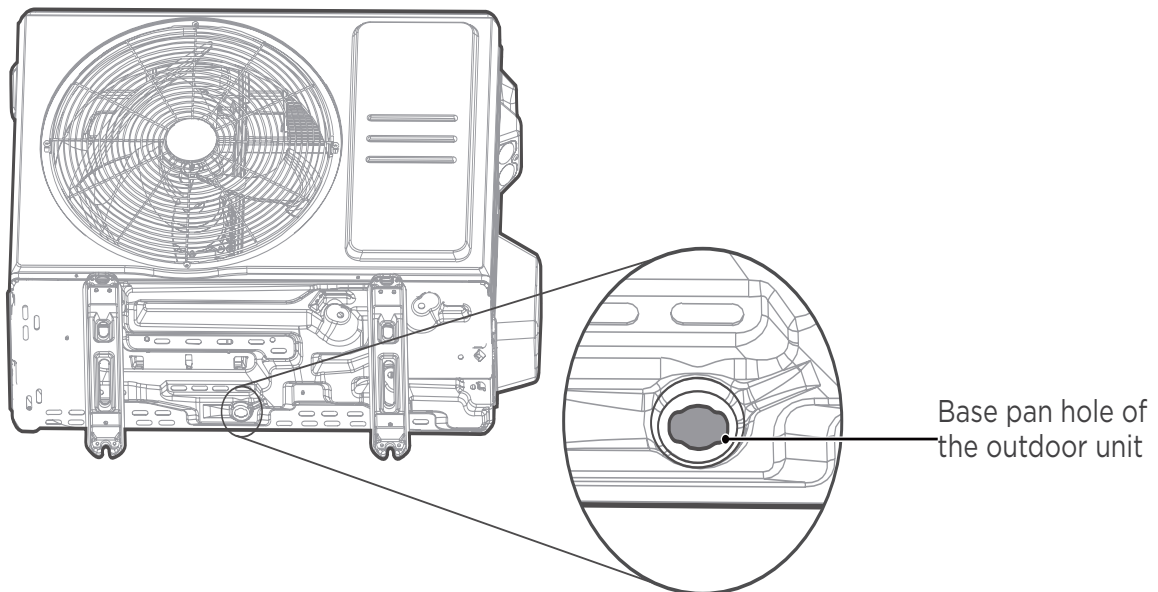
**DO NOT** install unit in the following locations:

- ⊘ Near an obstacle that will block air inlets and outlets
- ⊘ Near animals or plants that will be harmed by hot air discharge.
- ⊘ In a location that is exposed to large amounts of dust
- ⊘ Near a public street, crowded areas, or where noise from the unit will disturb others.
- ⊘ Near any source of combustible gas.
- ⊘ In a location exposed to a excessive amounts of salty air.

Step 2 - Install the Drain Joint

NOTE: Before bolting the outdoor unit in place, you must install the drain joint at the bottom of the unit. For units with a base pan built-in (with multiple holes for proper draining during defrost), the drain joint is not needed to be installed.

1. Locate the outdoor unit's base pan hole.

**Fig. 8 — Base pan hole**

2. Fit the rubber seal on the end of the drain joint that connects to the outdoor unit.

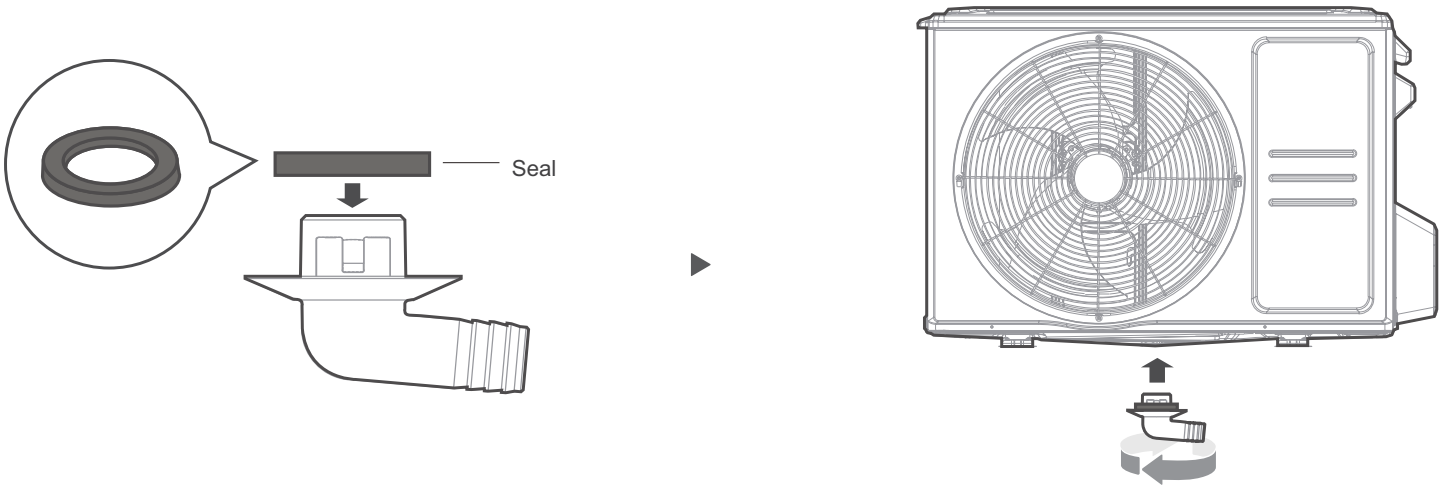


Fig. 9 — Fit the rubber seal

- 3. Insert the drain joint into the hole in the base pan of the unit. The drain joint will “click” in place.
- 4. Connect a drain hose extension (not included) to the drain joint to redirect water from the unit during heating mode.

NOTE: In cold climates, ensure the drain hose is as vertical as possible to ensure swift water drainage. If water drains too slowly, it can freeze in the hose and flood the unit.

Step 3 - Anchor the Outdoor Unit

WARNING

When drilling into concrete, eye protection is recommended at all times.

- The outdoor unit can be anchored to the ground or to a wall-mounted bracket with bolt (M10). Prepare the installation base of the unit according to Figure 10.
- Table 7 lists outdoor unit sizes and the distance between their mounting feet. Prepare the installation base of the unit according to the dimensions listed in Table 7.

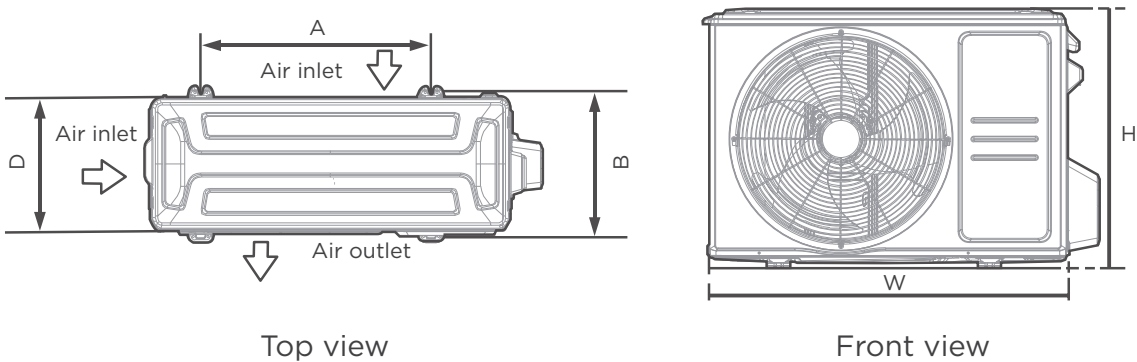


Fig. 10 — Anchor Outdoor unit

Table 7 — Dimensions

OUTDOOR UNIT DIMENSIONS W X H X D	MOUNTING DIMENSIONS	
	DISTANCE A	DISTANCE B
30.1inx 21.8inx 11.9in (765mmx555mmx303mm)	17.8in (452mm)	11.3in (286mm)
31.7inx 21.8inx 12.9in (805mmx554mmx330mm)	20.1in (511mm)	12.5in (317mm)
35.0inx 26.5inx 13.5in (890mmx673mmx342mm)	26.1in (663mm)	13.9in (354mm)
37.2inx 31.9inx 16.1in (946mmx810mmx410mm)	26.5in (673mm)	15.9in (403mm)

If you install the unit on the ground or on a concrete mounting platform, perform the following steps.

1. Mark the positions for four expansion bolts based on the dimensions chart.
2. Pre-drill holes for expansion bolts.
3. Place a nut on the end of each expansion bolt.
4. Hammer expansion bolts into the pre-drilled holes.
5. Remove the nuts from expansion bolts, and place the outdoor unit on bolts.
6. Put a washer on each expansion bolt, then replace the nuts.
7. Use a wrench to tighten each nut until snug.

If you install the unit on a wall-mounted bracket, perform the following steps.

1. Mark the position of bracket holes based on the dimensions chart.
2. Pre-drill the holes for the expansion bolts.
3. Place a washer and nut on the end of each expansion bolt.
4. Thread expansion bolts through holes in the mounting brackets, put mounting brackets in position, and hammer the expansion bolts into the wall.
5. Ensure the mounting brackets are level.
6. Carefully lift unit and place its mounting feet on brackets.
7. Bolt the unit firmly to the brackets.
8. If allowed, install the unit with rubber gaskets to reduce vibrations and noise.



CAUTION

Ensure the wall is made of solid brick, concrete, or of similarly strong material. The wall must be able to support at least four times the weight of the unit.

B. Refrigerant Connection Instructions

1. Run the interconnecting piping from the outdoor unit to the indoor unit.
2. Connect the refrigerant piping and drain line outside the indoor unit. Complete the pipe insulation at the flare connection then fasten the piping and wiring to the wall as required. Completely seal the hole in the wall.
3. Cut tubing to the correct length.
When preparing refrigerant pipes, take extra care to cut and flare them properly. This ensures efficient operation and minimizes the need for future maintenance.
 - a. Measure the distance between the indoor and outdoor units.
 - b. Using a pipe cutter, cut the pipe a little longer than the measured distance.
 - c. Make sure the pipe is cut at a perfect 90° angle.

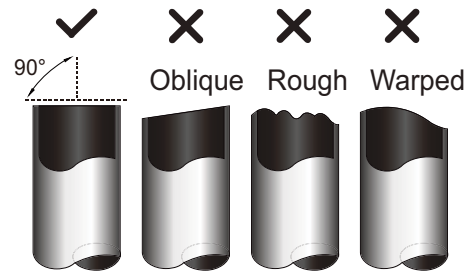


Fig. 11 — Pipe Cutting



CAUTION

DO NOT DEFORM PIPE WHILE CUTTING

Be extra careful not to damage, dent, or deform the pipe while cutting. This will drastically reduce the heating efficiency of the unit.

4. Remove Burrs

Burrs can affect the air-tight seal of the refrigerant piping connection. Therefore, they must be completely removed.

To remove:

- a. Hold the pipe at a downward angle to prevent burrs from falling into the pipe.
- b. Using a reamer or deburring tool, remove all burrs from the cut section of the pipe.

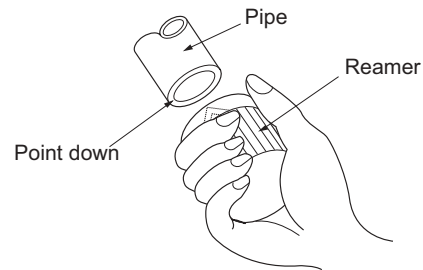


Fig. 12 — Deburring Tool

5. Flare Pipe Ends

Proper flaring is essential to achieving an airtight seal.

- a. After removing the burrs from the cut pipe, seal the ends with PVC tape to prevent foreign materials from entering the pipe.
- b. Sheath the pipe with insulating material.
- c. Place factory flare nut on pipe facing the proper direction. Make sure they are facing the right direction. Once the ends are flared, it is impossible to put them on or change their direction.

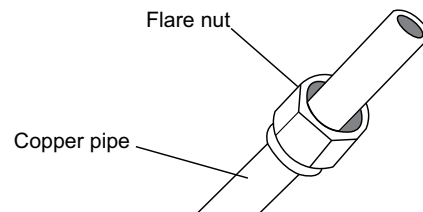


Fig. 13 — Copper pipe and flare nut

- d. Remove the PVC tape from ends of pipe when ready to perform the flaring work.

- e. Clamp the flare block on the end of the pipe. The end of the pipe must extend beyond the flare form.

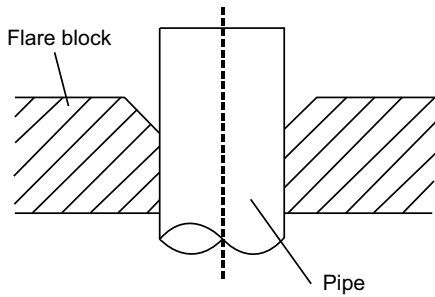


Fig. 14 — Flare Block

- f. Place the flaring tool onto the form.
g. Turn the handle of the flaring tool clockwise until the pipe is fully flared. Flare the pipe in accordance with the dimensions in Table 8.

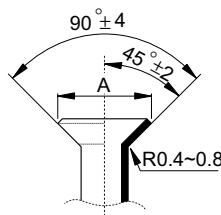


Fig. 15 — Flare Shape

Table 8 — Flare Nut Spacing

OUTER DIAMETER IN (MM)	A" IN (MM)	
	MIN.	MAX.
Ø 1/4in (Ø 6.35mm)	0.0275in (0.7mm)	0.05in (1.3mm)
Ø 3/8in (Ø 9.52mm)	0.04in (1.0mm)	0.063in (1.6mm)
Ø 1/2in (Ø 12.7mm)	0.04in (1.0mm)	0.07in (1.8mm)
Ø 5/8in (Ø 16mm)	0.078in (2.0mm)	0.086in (2.2mm)
Ø 3/4in (Ø 19mm)	0.078in (2.0mm)	0.094in (2.4mm)

- h. Remove the flaring tool and flare block, then inspect the end of the pipe for cracks and even flaring.
6. Connect the Pipes
Connect the copper pipes to the indoor unit first, then connect the pipes to the outdoor unit. Connect the low-pressure pipe first, then connect the high pressure pipe. When connecting the flare nuts, apply a thin coat of refrigeration oil to the flared ends of the pipes.
- a. Align the center of the two pipes that you will connect.

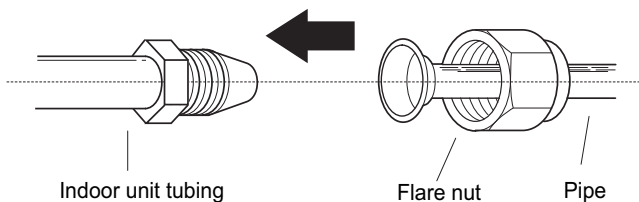


Fig. 16 — Align the center of the two pipes

- b. Tighten the flare nut as much as possible by hand.
c. Using a wrench, grip the nut on the unit tubing.
d. While firmly gripping the nut, use a torque wrench to tighten the flare nut according to the torque values listed in Table 9.

Table 9 — Tightening Torque

BRASS FLARE SIZES	RECOMMENDED SEATING TORQUE FOR BRASS FLARE NUTS		FLARE DIMENSIONS (A) (INCH/MM)	
	Ft-Lbs	N-M	Min	Max
1/4 (6.35)	13.3-14.7	18-20	0.33/8.4	0.34/8.7
3/8 (9.52)	23.6-28.8	32-39	0.52/13.2	0.53/13.5
1/2 (12.7)	36.1-42.8	49-59	0.64/16.2	0.65/16.5
5/8 (16)	42-52.4	57-71	0.76/19.2	0.78/19.7
3/4 (19)	49.4-74.5	67-101	0.91/23.2	0.93/23.7
7/8 (22)	62.7-81.1	85-110	1.04/26.4	1.06/26.9

NOTE: Use both a backup wrench and a torque wrench when connecting or disconnecting pipes to or from the unit.

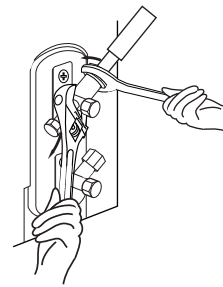


Fig. 17 — Torque wrench with backup wrench

CAUTION

Wrap insulation around the piping. Direct contact with the bare piping may result in burns or frostbite. Ensure the pipe is properly connected. Over tightening may damage the bell mouth and under tightening may lead to leakage.

All tubing bends should be performed with a properly sized tubing bender to prevent kinking or damaging the tubing.

- e. After connecting the copper pipes to the indoor unit, wrap the power cable, signal cable and the piping together with binding tape.

NOTE: While bundling these items together, DO NOT intertwine or cross the signal cable with any other wiring.

- f. Thread this lineset through the wall and connect it to the outdoor unit.
g. Insulate all piping, including the outdoor unit valves.

NOTE: DO NOT open the service valves until pressure test is complete.

7. Pressure Test Piping

**CAUTION**

Only use Dry Nitrogen to pressure test refrigerant systems. Use of other gases can result in injury, property damage or death.

NOTE: Use refrigeration gauges that are pressure rated for R454b refrigerant.

- Attach low side gauge hose to the 5/16" Schrader valve on the outdoor unit service valve.
- Attach the charging hose to the regulator on the dry nitrogen tank.
- Preset the nitrogen regulator to 550 psi.
- Slowly pressurize the line set until the low side gauge reads 500 psi. Do not exceed 550 psi.
- Close all the valves on the nitrogen tank and gauges.
- Allow the pressure test to stand for a minimum of 30 minutes.
- If the pressure holds, release the nitrogen and proceed. Review "E. Test Run" on page 21.
- If the pressure goes down in the 30 minute delay, leak check the tubing and flare fittings to identify the source of the leak. Return to Step C, above.

NOTE: Carefully bend the tubing in the middle according to the diagram below. DO NOT bend the tubing more than 90° or more than 3 times'

C. Wiring Instruction**Wiring Precautions**

WARNING - BEFORE PERFORMING ANY ELECTRICAL WORK, READ THESE WARNINGS.

All wiring must comply with local and national electrical codes, regulations and must be installed by a licensed electrician.

All electrical connections must be made according to the Electrical Connection Diagram located on the panels of the indoor and outdoor units.

If there is a serious safety issue with the power supply, stop work immediately. Explain your reasoning to the client, and refuse to install the unit until the safety issue is properly resolved.

Power voltage should be within 90-110% of rated voltage. (Insufficient power supply can cause malfunction, electrical shock, or fire.)

Installation of an external surge suppressor at the outdoor disconnect is recommended.

If connecting power to fixed wiring, a switch or circuit breaker that disconnects all poles and has a contact separation of at least 1/8in (3mm) must be incorporated in the fixed wiring. The qualified technician must use an approved circuit breaker or switch.

Only connect the unit to an individual branch circuit. Do not connect another appliance to that outlet.

Make sure to properly ground the air conditioner.

Every wire must be firmly connected. Loose wiring can cause the terminal to overheat, resulting in product malfunction and possible fire.

Do not let wires touch or rest against refrigerant tubing, the compressor, or any moving parts within the unit.

If the unit has an auxiliary electric heater, it must be installed at least 40in (1 m) away from any combustible materials.

To avoid getting an electric shock, never touch the electrical components soon after the power supply has been turned o. After turning o the power, always wait 10 minutes or more before you touch the electrical components.

Make sure that you do not cross your electrical wiring with your signal wiring.

This may cause distortion, interference or possibly damage to circuit boards.

No other equipment should be connected to the same power circuit.

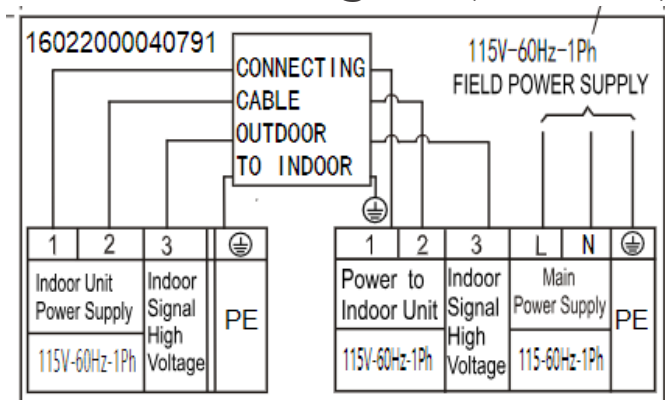
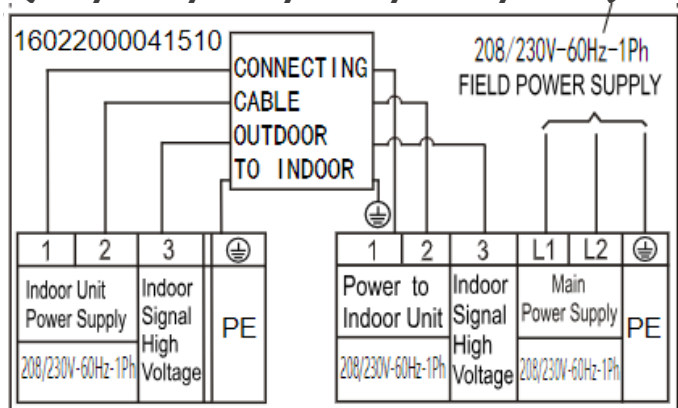
Connect the outdoor wires before connecting the indoor wires.

**WARNING**

Before performing any electrical or wiring work, turn off the main power to the system.

Connect Signal and Power Cables**WARNING**

All wiring work must be performed strictly in accordance with the wiring diagram located in the wire cover of the outdoor unit. Before performing any electrical or wiring work, turn off the main power to the system.

Connection Diagram**Connection Diagram (12K 115V)****Connection Diagram (9K/12K/18K/24K/30K/36K)****Fig. 18 — Connection Diagram****NOTES:**

- Do not use the thermostat wire for any connection between indoor and outdoor units.

2. All connections between the indoor and outdoor units must be made as shown in Figure 18. The connections are sensitive to polarity and will result in a fault code.

The outside unit's terminal block is protected by an electrical wiring cover on the side of the unit. A comprehensive wiring diagram is printed on the inside of the wiring cover.

1. Remove the wire cover from the unit by loosening the 3 screws.
2. Dismount the caps on the conduit panel.
3. Temporarily mount the conduit tubes (not included) on the conduit panel.
4. Properly connect both the power supply and low voltage lines to the corresponding terminals on the terminal block.
5. Ground the unit in accordance with local codes.
6. Be sure to size each wire allowing several inches longer than the required length for wiring.
7. Use lock nuts to secure the conduit tubes.

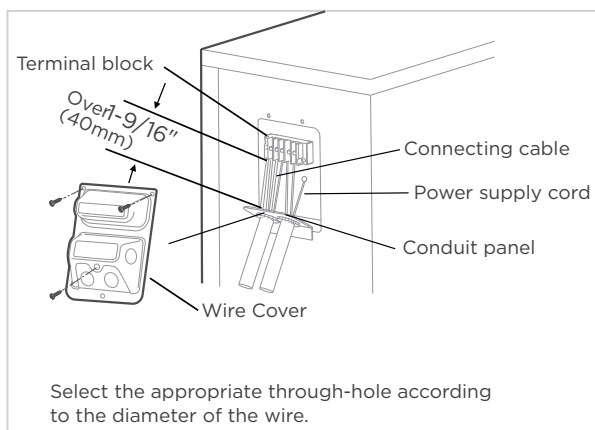


Fig. 19 — Select the through-hole

Properly Connect the Wire Lines

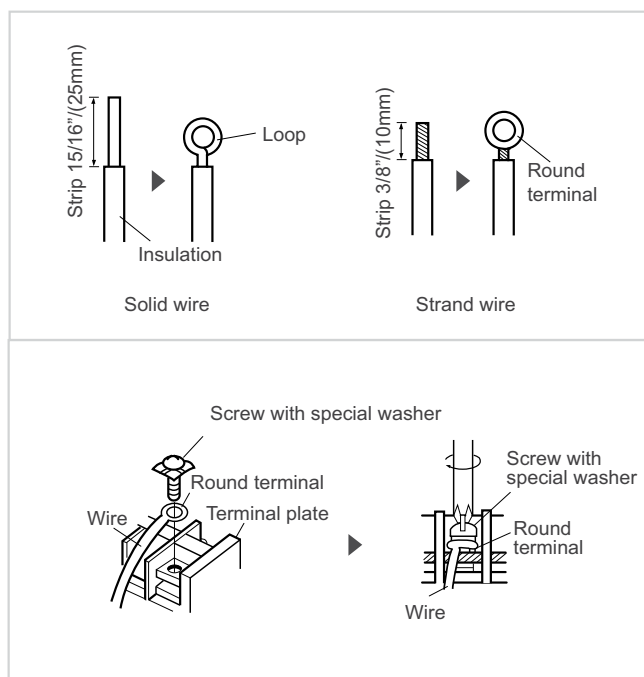


Fig. 20 — Properly Connect the Wire Lines

1. The treatment about the end of the wire.
2. Connect the line to the corresponding terminals on the terminal block.

WIRING

All wires must be sized per NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use Electrical Data table MCA (minimum circuit amps) and MOCP (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

Power and Communication Wiring: The main power is supplied to the outdoor unit. The field supplied 14/3 power/communication wiring, from the outdoor unit to the indoor unit, consists of four (4) wires and provides the power for the indoor unit. Two wires are high voltage AC power, one is communication wiring and the other is a ground wire.

To minimize communication interference: If installed in a high Electromagnetic field (EMF) area and communication issues arise, a 14/2 stranded shielded wire can be used to replace 2 (L2/N) and 3 (S) between the outdoor and indoor units - landing the shield onto the ground in the outdoor unit only.

Table 10 — Wiring Sizes

CABLE	CABLE SIZE	REMARKS
Connection Cable	14AWG	3 wire + Ground 1Φ 208/230 V (Stranded wire is recommended)



WARNING

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Wires should be sized based on NEDC and local codes.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Be sure to comply with local codes while running wire from the indoor unit to the outdoor unit. Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Ensure all wiring is tightly connected.

No wire should touch the refrigerant tubing, compressor or any moving parts. Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner. Connecting cable with conduit shall be routed through the hole in the conduit panel.

Electrical and Gas Leaks



WARNING

BEFORE TEST RUN

Only perform test run after you have completed the following steps:

Electrical Safety Checks – Confirm that the unit's electrical system is safe and operating properly

Gas Leak Checks – Check all flare nut connections and confirm that the system is not leaking

Confirm that gas and liquid (high and low pressure) valves are fully open

ELECTRICAL DATA**Table 11 — Electrical Data**

OUTDOOR UNIT		12K 115V	09K	12K	18K	24K	30K	36K
		(115V)	(208/230V)	(208/230V)	(208/230V)	(208/230V)	(208/230V)	(208/230V)
MINIMUM CIRCUIT AMPACITY (MCA)	A	18	12	12	18	24	28	33
MOP	A	20	15	15	20	25	30	35
VOLTAGE-PHASE-FREQUENCY		115-1-60	208/230-1-60					
MAX – MIN VOLTAGE RANGE		127-104	253-187					
COOLING								
RUNNING CURRENT	(A)	10.3	4.7	4.1	6.3	7.4	11.2	18
POWER CONSUMPTION	(W)	923	604	916	1406	1714	2500	4138
HEATING								
RUNNING CURRENT RANGE	(A)	10.5	3,3	4	6.7	8.3	13.1	16.2
POWER CONSUMPTION	(W)	950	750	902	1507	1852	2857	3700

*Permissible limits of the voltage range at which the unit will operate satisfactorily.

LEGEND

FLA - Full Load Amps

MCA - Minimum Circuit Amps

MOP - Maximum Overcurrent Protection

D. Air Evacuation**EVACUATION COIL AND TUBING SYSTEM**



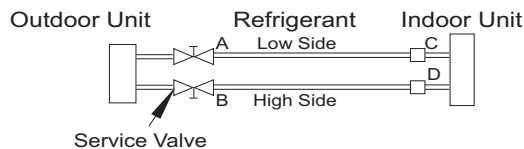
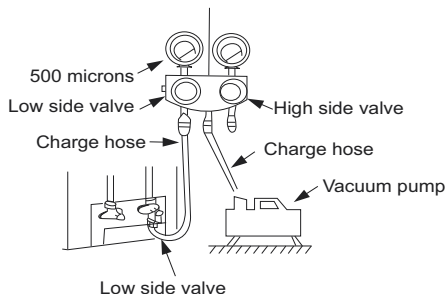
CAUTION

UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Never use the system compressor as a vacuum pump.

Refrigerant tubes and the indoor coil should be evacuated using the recommended 500 microns deep vacuum method. The alternate triple evacuation method may be used if the procedure outlined below is followed.

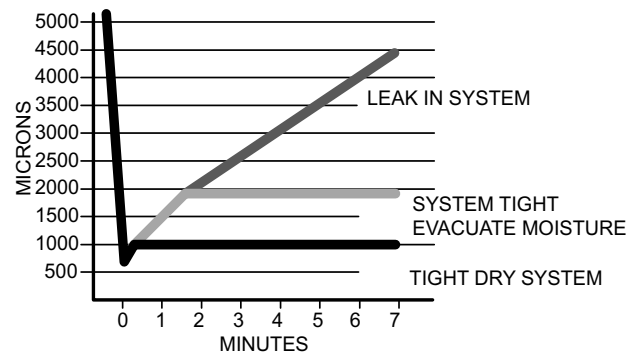
NOTE: Always break a vacuum with dry nitrogen.

**Fig. 21 — Service Valve****Fig. 22 — Manifold****Evacuation**

Evacuation of the system removes air or nitrogen (non-condensables) as well as moisture. A proper vacuum assures a tight, dry system before charging with refrigerant. The two methods used to evacuate a system are the Deep Vacuum Method and the Triple Evacuation Method.

Deep Vacuum Method

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 microns and a vacuum gauge capable of accurately measuring this vacuum depth. The deep vacuum method is the most positive way of assuring a system is free of air and moisture (see Fig. 23).

**Fig. 23 — Deep Vacuum Graph****Triple Evacuation Method**

The triple evacuation method should be used when the vacuum pump is not capable of pumping down to 500 microns and system does not contain any liquid water. Refer to Fig. 24 and proceed as follows:

1. Attach refrigeration gauges and evacuate system down to 28 in. of mercury and allow pump to continue operating for an additional 15 minutes.
2. Close service valves and shut off vacuum pump.
3. Connect a nitrogen cylinder and regulator to system and flow nitrogen until system pressure is 2 psig.

4. Close service valve and allow system to stand for 1 hour. During this time, dry nitrogen will be able to diffuse throughout the system absorbing moisture.
5. Repeat this procedure as indicated in Fig. 24. System will then be free of any contaminants and water vapor.

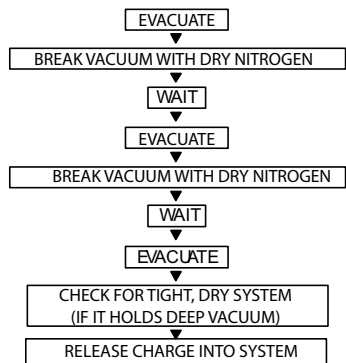


Fig. 24 — Triple Evacuation Method

Final Tubing Check

Important: Check to be certain factory tubing on both the indoor and outdoor unit has not shifted during shipment. Ensure tubes are not rubbing against each other or any sheet metal. Pay close attention to the feeder tubes, making sure wire ties on feeder tubes are secure and tight.

E. Test Run

CAUTION

Failure to perform the test run may result in unit damage, property damage, or personal injury.

Before Test Run

A test run must be performed after the entire system has been completely installed. Confirm the following points before performing the test:

- a) Indoor and outdoor units are properly installed.
- b) Piping and wiring are properly connected.
- c) No obstacles near the inlet and outlet of the unit that might cause poor performance or product malfunction.
- d) Refrigeration system does not leak.
- e) Drainage system is unimpeded and draining to a safe location.
- f) Heating insulation is properly installed.
- g) Grounding wires are properly connected.
- h) Length of the piping and additional refrigerant capacity have been recorded.
- i) Power voltage is the correct voltage for the air conditioner

Run Test Instructions

1. Open both the liquid and gas stop valves.
2. Turn on the main power switch and allow the unit to warm up.
3. Set the air conditioner to COOL mode.
4. For the Indoor Unit
 - a. Ensure the remote control and its buttons work properly.
 - b. Ensure the louvers move properly and can be changed using the remote control.
 - c. Double check to see if the room temperature is being registered correctly.
 - d. Ensure the indicators on the remote control and the display panel on

the indoor unit work properly.

- e. Ensure the manual buttons on the indoor unit works properly.
 - f. Check to see that the drainage system is unimpeded and draining smoothly.
 - g. Ensure there is no vibration or abnormal noise during operation.
5. For the Outdoor Unit
- a. Check to see if the refrigeration system is leaking.
 - b. Make sure there is no vibration or abnormal noise during operation.
 - c. Ensure the wind, noise, and water generated by the unit do not disturb your neighbors or pose a safety hazard.
6. Drainage Test
- a. Ensure the drainpipe flows smoothly. New buildings should perform this test before finishing the ceiling.
 - b. Remove the test cover. Add 2,000ml of water to the tank through the attached tube.
 - c. Turn on the main power switch and run the air conditioner in COOL mode.
 - d. Listen to the sound of the drain pump to see if it makes any unusual noises.
 - e. Check to see that the water is discharged. It may take up to one minute before the unit begins to drain depending on the drainpipe.
 - f. Make sure that there are no leaks in any of the piping.
 - g. Stop the air conditioner. Turn o the main power switch and reinstall the test cover.

NOTE: If the unit malfunctions or does not operate according to your expectations, please refer to the Troubleshooting section of the Owner's Manual before calling customer service.

CARE AND MAINTENANCE

To help ensure high performance and minimize possible equipment failure, periodic maintenance must be performed on this equipment. Maintenance frequency may vary depending upon geographic areas.

TROUBLESHOOTING

For ease of service, the systems are equipped with diagnostic code display LEDs on both the indoor and outdoor units. The outdoor diagnostic display has one red LED on the outdoor unit board and is limited to very few errors. The indoor diagnostic display is a combination of flashing LEDs on the display panel or the front of the unit.

There may be a few error codes displayed in the indoor unit that might relate to the outdoor unit's problems. If possible, always check the diagnostic codes displayed on the indoor unit first.

ERROR CODES

Table 12 — Error Codes

Display	Malfunction and Protection Indication	Display	Malfunction and Protection Indication
EC01	Other indoor unit refrigerant sensor detects a leak (multi-zone)	PC08	ODL Current Protection
EC07	Outdoor DC fan motor speed out of control	PC10	ODL AC voltage too low protection
EC51	ODU EEPROM error	PC11	ODL DC bus voltage too high protection
EC52	Condenser coil temperature sensor (T3) error	PC12	ODL DC bus voltage too low protection
EC53	ODU temperature sensor (T4) error	PC30	System pressure overload protection
EC54	ODU exhaust temperature sensor error	PC31	System pressure too low protection
EC55	ODU IPM module temperature sensor error	PC40	Communication failure between outdoor main control chip and the driver chip
EC56	ODU T2B sensor	PC41	Compressor current sampling circuit failure
EH00	IDU EEPROM error	PC42	Compressor starting failure
EH0A	Indoor EEPROM Parameter error	PC43	Compressor lost phrase protection
EH01	Refrigerant Sensor detects a leak	PC44	Compressor zero speed protection
EH02	Working condition of the refrigerant sensor is out of range and a leak is detected	PC45	Voltage drop
EH03	Working condition of the refrigerant sensor is out of range	PC46	Compressor speed out of control
EH02	Zero-crossing signal detection error	PC49	Compressor over current error
EH03	Indoor fan motor speed is out of control	PC0A	Condenser high temperature protection
EH31	Protection for low DC bus voltage of the external fan	PC0F	PFC failure
EH32	Protection for high DC bus voltage of the external fan	PC0L	Outdoor low temperature protection
EH60	IDU ENV temperature T1 sensor error	PH09	IDU anti-cold wind stop machine
EH61	IDU pipe temperature T2 sensor error	PH90	Evaporator high temperature protection
EH0b	IDU PCB and display communication error	PH91	Evaporator low temperature protection
FH0C	Indoor Unit humidity sensor malfunction	LC01	Condenser high temperature frequency limited (L1)
FHCC	Refrigerant Sensor Error	LC02	Compressor Discharge Pipe High temperature frequency limited (L2)
EL01	IDU and ODU Communication Error	LC03	Current frequency limited (L3)
EL0C	System leaks refrigerant	LC05	Voltage frequency limited (L5)
FH0P	Wireless Module Self-Test Failure	LC06	IPM module temperature frequency limited
FL09	New and old platform mismatch failure	LH00	Evaporator temperature frequency limited (L0)
PC00	ODU IPM Protection	LH07	Remote Control frequency limitation in effect
PC01	ODU Voltage Protection		
PC02	Compressor top temperature (IPM module temperature protection)	---	Mode conflict fault
		nA	No fault or protection
PC03	System Pressure Protection		

Table 13 — Refrigerant Leak Detection Error Codes

FHCC	Refrigerant Sensor Error
EL01	IDU and ODU Communication Error
EL0C	System Leaks Refrigerant

If you receive one of the codes in Table 13, call a technician as soon as possible. No need to panic, the unit goes into TURBO mode until the error code is cleared. There is a “beep” noise coming from the indoor unit, which is normal in this case.

DUCTLESS START-UP CHECKLIST - Single Zone

Installation Data

Site Address: _____

City: _____ State: _____ Zip Code: _____

Installing Contractor: _____ Contractor Contact #: () _____ - _____

Job Name: _____ Start-up Date: _____

Distributor: _____

System Details

UNITS	MODEL NO.	SERIAL NO.	CONTROLLER
OUTDOOR UNIT			
INDOOR UNIT A			

Are the outdoor unit and indoor unit compatible?

YES: _____ NO: _____

Wiring Electrical

Wire Size and Type Used? AWG: _____ TYPE: _____

Are there any breaks, splices, wire nuts or butt connectors between the outdoor unit and the indoor unit?

YES: _____ NO: _____

Was the wiring from the outdoor unit port to the correct indoor unit verified?

YES: _____ NO: _____

REMARKS: _____

Voltage Check

Wiring: Single Zone

Outdoor Unit Disconnect	1(L1):GND		Outdoor Unit Terminal Block	1(L1):GND		NOTES: _____ _____ _____ _____
	2(L2):GND			2(L2):GND		
	1(L1):L2(2)			1(L1):2(L2)		
Indoor Unit Voltage Check @ Outdoor Unit	1(L1):GND		Indoor Unit Voltage Check @ Indoor Unit	1(L1):GND		NOTES: _____ _____ _____ _____
	2(L2):GND			2(L2):GND		
	1(L1):2(L2)			1(L1):2(L2)		
	2(L2):3(S)			2(L2):3(S)		

Outdoor Unit Disconnect	1(L1):GND		Outdoor Unit Terminal Block	1(L1):GND		NOTES: _____ _____ _____ _____
	2(L2):GND			2(L2):GND		
	1(L1):L2(2)			1(L1):2(L2)		
Indoor Unit Voltage Check @ Outdoor Unit	1(L1):GND		Indoor Unit Voltage Check @ Indoor Unit	1(L1):GND		NOTES: _____ _____ _____ _____
	2(L2):GND			2(L2):GND		
	1(L1):2(L2)			1(L1):2(L2)		
	2(L2):3(S)			2(L2):3(S)		

Ductless Start-Up Checklist (CONT)

Piping

Leak Check:

System held 500 psig (max. 550psi) for a minimum of 30 minutes using dry nitrogen. YES: _____ NO: _____

Evacuation Method:

- Was the Triple Evacuation Method used as outlined in the installation manual? YES: _____ NO: _____
- Was the Deep Vacuum Method used as outlined in the installation manual? YES: _____ NO: _____
- Did the System Hold 500 microns for 1 hour? YES: _____ NO: _____
- Does the line set match the diameter of the evaporator connections? YES: _____ NO: _____
- For Conventional Fan Coils, does the line set match the outdoor unit size? YES: _____ NO: _____

Single Zone Piping:

Has the liquid pipe length been measured and the additional charge calculated? Size: _____ Length: _____ Charge: _____

NOTES: _____

NOTE: Final Charge Amount must be recorded!

PORT	LIQUID SIZE		SUCTION SIZE		LENGTH	CHARGE	NOTES: _____
A							_____

Performance Check

For 1:1 Single Zone Systems: Adjust the set-point to create an operational call for the desired testing operation. Allow the system to run for a minimum of 10 min. and record the following details:

(Operational data recorded on applicable heads with the wireless remote controller's Point Check function)

UNIT	SET-POINT	MODE	T1	T2	T3	T4	Tb	Tp	Th	LA/Lr
A										

NOTE:

- T1 - Ambient Space Temperature Sensor
- T2 - IDU Coil Temperature Sensor
- T3 - Outdoor Coil Temperature Sensor
- T4 - Outdoor Ambient Temperature
- Tb - Suction Line Temperature @PMV
- Tp - Discharge Temperature Sensor
- Th - IPM Board Temperature
- LA/Lr - PMV Temperature

Error Codes

Were there any error codes present at start-up? YES: _____ NO: _____

INDOOR UNIT ERROR CODE	NOTES:
OUTDOOR UNIT ERROR CODE	
WALL CONTROLLER	
24V INTERFERENCE	

Comments:
